

LACHLAN KNAPP

Physics and Applied Mathematics Student • Minor in Computer Science

Urbana-Champaign, IL | knapplachlan@gmail.com | lgknapp@illinois.edu |
(224) 813-2324

EDUCATION

University of Illinois Urbana-Champaign

2025 – Present

Bachelor of Science in Physics and Bachelor of Science in Liberal Arts & Sciences in Applied Mathematics

GPA:
3.84/4.0

Minor in Computer Science

Relevant Coursework: Ordinary Differential Equations, Data Structures and Algorithms, Introduction to Computational Physics, Relativity and Math Applications, Classical Mechanics I

EXPERIENCE

Undergraduate Researcher - Chemical Imaging and Structures Laboratory

2026 – Present

Beckman Institute for Advanced Science and Technology

- Modeled AFM-IR systems in 2D axisymmetric and full 3D coordinate systems with COMSOL and SOLIDWORKS for analyzing and improving AFM tip designs for dye-free chemical imaging and cancer research applications.
- Operated environmental scanning electron microscopes to conduct AFM-IR probe damage analysis and optimization.
- Experience working in a deadline-driven lab environment collaborating with researchers and scientists from various different fields.
- Presented Research at the STEM Career Exploration and Symposium poster session in July, 2026.

Club President

2024 – 2025

Highland Park High School Lettuce Eating Club

- Raised over \$500 for charity during my high school's charity drive through a succulent sale.
- Created a welcoming and fun environment focused on promoting healthy eating and mindfulness about the environment.

Caddie

2021 – 2025

Exmoor Country Club

- Carried bags, cleaned clubs, gave advice and socialized with club members.
- Worked in a competitive environment requiring extensive golf knowledge.
- Managed various professional social engagements with clients.

PROJECTS

Modeling Chaotic Dynamics of a Double Pendulum using JAX

2026

- Used Python to model a double pendulum with Lagrangian mechanics using NumPy to optimize runtime.
- Explored different neural network training methods from standard Stochastic Gradient Descent (SGD) to Adaptive Moment Estimation (ADAM) using JAX for training.
- Generated trajectory animations and loss visualizations using Matplotlib to evaluate convergence.

SKILLS

Languages & Libraries: Java, C++, Python, NumPy, JAX, Matplotlib | **Modeling & CAD:** COMSOL, SOLIDWORKS, Fusion, Blender | **Tools & Web:** Git, Docker, HTML, CSS